$$
\begin{array}{r|l}
 & 5x-2 \\
\hline
x-1 & 5x^2-7x+2 \\
 & 5x^2-5x \\
 & (-)\ \ (+) \\
\hline
 & -2x+2 \\
 & -2x+2 \\
 & (+)\ \ (-) \\
\hline
 & 0
\end{array}
$$

(i) $\frac{5x^2}{x} = 5x$

(ii) $5x(x-1) = 5x^2-5x$

(iii) $-\frac{2x}{x} = -2$

(iv) $-2(x-1) = (-2x+2)$

$\therefore$ Quotient $= 5x-2$ Remainder $= 0$

Example 3.34 Find quotient and the remainder when $f(x)$ is divided by $g(x)$

(i) $f(x) = (8x^3-6x^2+15x-7)$, $g(x) = 2x+1$. (ii) $f(x) = x^4-3x^3+5x^2-7$, $g(x) = x^2+x+1$

Solution

(i) $f(x) = (8x^3-6x^2+15x-7)$, $g(x) = 2x+1$

$$
\begin{array}{r|l}
 & 4x^2-5x+10 \\
\hline
2x+1 & 8x^3-6x^2+15x-7 \\
 & 8x^3+4x^2 \\
 & (-)\ \ (-) \\
\hline
 & -10x^2+15x \\
 & -10x^2-5x \\
 & (+)\ \ (+) \\
\hline
 & 20x-7 \\
 & 20x+10 \\
 & (-)\ \ (-) \\
\hline
 & -17
\end{array}
$$

$\therefore$ Quotient $= 4x^2-5x+10$ and Remainder $= -17$

(ii) $f(x) = x^4-3x^3+5x^2-7$, $g(x) = x^2+x+1$

$$
\begin{array}{r|l}
 & x^2-4x+8 \\
\hline
x^2+x+1 & x^4-3x^3+5x^2+0x-7 \\
 & x^4+x^3+x^2 \\
 & (-)\ \ (-)\ \ (-) \\
\hline
 & -4x^3+4x^2+0x \\
 & -4x^3-4x^2-4x \\
 & (+)\ \ (+)\ \ (+) \\
\hline
 & 8x^2+4x-7 \\
 & 8x^2+8x+8 \\
 & (-)\ \ (-)\ \ (-) \\
\hline
 & -4x-15
\end{array}
$$

$\therefore$ Quotient $= x^2-4x+8$ and Remainder $= -4x-15$

3.6.2 Synthetic Division

Synthetic Division is a shortcut method of polynomial division. The advantage of synthetic division is that it allows one to calculate without writing variables, than long division.

Example 3.35 Find the quotient and remainder when $p(x) = (3x^3-2x^2-5+7x)$ is divided by $d(x) = x+3$ using synthetic division.

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